

AISI CHALLENGE FUND

PRIORITY RESEARCH AREAS

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Purpose of this document

This document is intended for **prospective applicants** to the AISI Challenge Fund. It outlines four priority research areas: **Safeguards, Control, Alignment**, and **Societal Resilience**. For each, we provide an overview of key research challenges and directions that AISI is looking to progress.

The Challenge Fund is designed to support a diverse range of projects that contribute to safe and secure AI development. While we are particularly interested in proposals that align with our priority research areas, we also welcome proposals that explore other innovative topics relevant to the safe and secure development of AI systems.

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SAFEGUARDS

Summary

As AI systems become more capable and integrated into the economy, they will increasingly be targeted by adversaries looking to take advantage of their weaknesses. This includes adversaries misusing AI systems to aid in perpetrating large-scale harms, or adversaries disrupting operation of deployed AI systems to cause data loss or damage to critical systems.

Research that seeks to understand, evaluate, and improve the technical measures designed to address these risks (“safeguards”) is urgently needed.

Key Research Challenges

- **Defending hosted frontier AI systems against misuse.** When AI systems are hosted by benign actors, a range of safeguards can be used to prevent adversaries using the AI to aid in illegal activity. These can include *model-level* safeguards like safety training; *system-level* safeguards like real-time monitors; *access* safeguards like giving vetted users or organisations access to less safeguarded or more advanced systems; or *maintenance* safeguards like rapid vulnerability remediation. The challenge also includes alternative access forms, like mitigating against misuse by insiders and defending fine-tuning APIs. A key difficulty is developing safeguards that prevent genuinely malicious activity whilst remaining cost-effective appropriate to the level of risk, and not significantly dampening the capability or utility of the model.
- **Defending against 3rd party attacks.** Exposure to attacker-controlled data during training (“data poisoning”) or inference (“prompt-injection”)—as well as direct adversarial manipulation of model weights (“model poisoning”)—can lead to adversaries controlling the actions or goals of otherwise benign AI systems. This could lead to AI agents exfiltrating sensitive information or causing large amounts of harm, or broad attacks on the availability of critical AI systems.
- **Mitigating misuse of open-weight models.** Open-weight models enable a range of defensive and beneficial use-cases. However, when AI model weights are made public, system-level safeguards, access safeguards, and maintenance safeguards are made largely infeasible; and model-level safeguards are made much more difficult, as users can fine-tune away refusal behaviour or design attacks in a full-information (white-box) setting. Developing reliable methods for preventing or mitigating risks from misuse of open-weight models remains an open problem.

SAFEGUARDS

Potential Research Directions

- **Improving safeguard evaluations.** We are interested in realistic evaluation environments for safeguards across all the challenges above, particularly those which assess autonomous agents acting in real-world environments and include careful measurements of costs and user friction. These include better evaluations of misuse safeguards like unlearning, rapid vulnerability remediation, and asynchronous monitoring, as well as poisoning and prompt injection defences. Along with evaluation environments, we would like to see work that develops stronger attacks to stress-test defences—especially those that can be incorporated into defensive measures. We'd also be interested to collaborate and fund work that understands fundamental limitations in specific defences or classes of defence.
- **Improving defences.** Specifically, work in addressing deficiencies in current safeguards, such as designing more resilient unlearning or data-filtering methods that remove a narrow harmful capability without degrading other model capabilities; developing better detection or mitigation strategies against data or model poisoning; or improving monitoring systems to cheaply and effectively catch instances of harm occurring, especially in agentic settings.
- **Longer-term bets.** We would also like to see research which explores directions that are less likely to provide short-term impact but will lead to substantial progress if successful. As a few examples, researchers could explore pre-training or architecture-level defences, design new paradigms for post-hoc safety training or unlearning or explore architectural changes to separate trusted and untrusted data sources to prevent prompt injection.

CONTROL

Summary

Most current safeguards for catching AI's unsafe behaviours are not designed to resist intentional subversion by misaligned AIs. Furthermore, current alignment methods can't guarantee that an AI's goals and actions match human intention.

[AI control](#) aims to design and evaluate *control protocols*. *Control protocols* are plans designed to prevent unsafe actions by AI systems, even if those AIs were misaligned and intended to subvert the safeguards. Promising protocols vary widely, from “*only deploy your AI systems inside a cyber-secure environment*” to “*use another AI to monitor all of your AI's actions and get a human to audit the most suspicious behaviour*”.

Control evaluations attempt to measure the safety of protocols. A core challenge here is experimental design: how can we estimate safety without access to real-world deployments, or real subversive models? Existing control evaluations use simplified environments as proxies for real-world deployments, human-AI red-teams as proxies for subversive models, and statistical modelling to estimate the relative risk of deploying at scale.

Key Research Challenges

Existing research has prioritised one of the most critical environments – AIs deployed within frontier AI developers' systems. Soon highly autonomous and capable AIs may be deployed within such environments, with access to sensitive code and infrastructure. To ensure such deployments are safe, we need protocols that have been shown to be adversarially robust in similar settings. Only a handful of AI control evaluations have been conducted so far, and there are many challenges to scaling control to meet the demands of future deployments:

- **Creating realistic proxy environments.** Designing experimental environments that are as similar to the real deployments as possible is challenging. For example, closely replicating the infrastructure and codebases used by leading developers.
- **Creating realistic red-team AIs.** The accuracy of evaluations depends heavily on the red-teaming AI used to stress-test the protocol. Different implementations of red-team AIs balance different levels of realism, conservatism and practicality.
- **Understanding human-in-the-loop protocols.** Some protocols rely on humans to audit the most suspicious actions from AI systems. Understanding and augmenting human's ability to investigate LLM behaviour's is essential, especially as models achieve superhuman performance on some tasks.
- **Relaxing assumptions.** Existing control evaluations make many assumptions that may not hold in real deployments. For example: that AIs are either always subversive or never subversive; or that models are either trusted or untrusted.

CONTROL

- **Securing AI developer infrastructure against insider threats from AI systems.** Preventing AI's from taking dangerous actions will be easier with stronger traditional information security protocols in place, ideally codesigned with control protocols.
- **Develop alternative control protocols.** Prior work has proposed a number of control protocols. However, improvements may be necessary, especially if current protocols fail in more realistic and complex environments, with less restrictive assumptions.
- **Scaling protocols to (and beyond?) AGI.** Prior evaluations consider AIs performing tasks at a sub-human level. It remains an open-question how far control protocols could scale to AI systems performing tasks at a superhuman level.

Potential Research Directions

- **Create more realistic red-team AIs.** [Prior work](#) on control has stress-tested protocols against language-models utilising a human-optimised scaffold, unseen by the protocol. A more realistic evaluation could, for example, modify the language model's weights, [using finetuning](#) or [transformer-compiled cryptographic algorithms](#).
- **Understand human-in-the-loop protocols.** Whereas previous research has focused on using LLMs as proxies for human auditing, we are excited to see empirical research on real human auditors. This could involve investigating how effective human auditors are against both LLM-generated and human-generated code vulnerabilities. We would also be excited about research into the factors that determine the success of human auditing (e.g., existing skillsets, time taken per trajectory and additional tooling provided).
- **Study analogous security measures for human insider threats.** Traditional software engineering security protocols might be adaptable to mitigate risks posed by AI agents acting as insider threats, similar to human threats. However, there is limited research at the intersection of AI control and conventional security strategies.
- **Develop alternative control protocols.** Potential future protocols could include white-box techniques such as linear probes, ensembles of monitors, hierarchical auditing and specific anti-collusion mitigations (such as better paraphrasing, or activation steering).
- **Investigating scaling protocols to (and beyond) AGI.** We're interested in work that convincingly argues that control can or cannot scale to super-human systems, across a wide range of domains.

ALIGNMENT

Summary

AI alignment is a field of research focused on ensuring that an AI's goals and actions match human intention. Misalignment occurs when an AI's behaviour deviates from these intentions, potentially causing harm. This might happen for various reasons, such as a disconnect between objectives used during training and the complexities of a real-world application, or when opaque internal reasoning makes it hard to predict any unintended outcomes. As AI systems become increasingly capable, methods to oversee or control their behaviour are likely to become less reliable, necessitating the need for robust 'AI alignment' methods to prevent substantial harm.

Key Research Challenges

Existing work on alignment has been focused on empirical iteration: improving over previous models on average-case benchmark performance. Reorienting the field to prepare for AGI deployment requires developing both methods and evaluations suitable for models deployed on high-stakes tasks—including tasks going beyond human ability.

- **Alignment methods:** We are interested in improvements over existing methods (listed on the next page) as well as developing novel alignment methods. Alignment methods must address three problems:
 - **Scalable oversight:** How can we correctly reward desired behaviours beyond humans' ability to efficiently judge them?
 - **Interpretability for worst-case guarantees:** Given error bounds from training, how do we ensure that random errors do not have correlations and structure which will be used to subvert human intent when deployed?
 - **Exploration and elicitation:** To train AIs on tasks where human supervision is unavailable, we expect to use AI assistance to help humans supervise AI systems. One way to mitigate unforeseen errors by AI systems is to elicit the AI's [maximum performance](#) at the assistance task.
- **Alignment evaluations:** Measuring the effectiveness of alignment methods requires developing evaluations without relying on human annotation. We are interested in development of new evaluation methodologies modelling imperfect human judgment e.g. via [partial observability](#), [sandwiching](#), or [adversarial initialisations](#).
- **Automated alignment:** Many researchers and AI developers are hoping to partially automate the search for and evaluation of alignment algorithms. We believe this may be feasible for some subproblems of alignment, but that obstacles such as [automation collapse](#) may block this approach on other subproblems. We are interested in research that maps or expands the set of subproblems that can be tackled, such as by studying long-horizon generalisation, reward hacking in the code agent setting, etc.

ALIGNMENT

Potential Research Directions and Existing Work

We are excited to see work extending and critiquing existing work on alignment methods. This includes:

Alignment methods:

- **Scalable oversight:**
 - **Debate:** [theoretical](#) and [empirical](#) work on recursive decomposition of claims assuming the human supervisor has limited budget to verify claims. Progress is needed on minimising theoretical assumptions, and on verifying empirically that methods are robust to contingencies such as data quality or judge competence.
 - **Faithful reasoning:** [measurement](#) and [methods](#) for verifying that model responses reflect claimed justifications given in chains of thought. Future work may consider relating [causal](#) interventions to faithfulness.
- **Interpretability and worst-case guarantees:**
 - [Tail risk estimation](#) for rare model behaviours. Open problems on theory have been [identified by ARC](#).
 - [Proving bounds on model behaviour](#) via interpretability.

Tools for alignment: Ensuring the adequacy of alignment methods requires an understanding of the details of training and inference. In the former case, we would like to understand training dynamics in neural networks, and in the latter case we would like to model scaffolding and deployment contexts. As examples of work in this area we have:

- [Singular learning theory](#) (SLT): SLT provides a Bayesian analysis relevant to neural network generalisation which can be used to analyse generalization behaviour e.g. via [training dynamics](#).
- [Scientist AIs](#): To simplify the alignment problem by targeting AIs exclusively at scientific problems, this agenda involves scoping training methods which are appropriate to this context and can offer improved alignment guarantees.

Novel theoretical frameworks and problem statements: The precise problem specification which alignment methods must satisfy is not yet well defined. We are interested in work which formalises previously identified problems, and which identifies novel problem areas. Examples of existing work here includes:

- [Partial observability](#) in reward modelling. This may be extended to cover computational limitations to the overseer.
- [Alignment faking](#): ensuring models conform given updated or improved training objectives, c.f. also related work on [incomplete preferences](#).
- [Scheming](#): formalising conditions under which a model can strategically identify differences in training and deployment distributions without detection. This has been studied in [the backdoors setting](#).

SOCIETAL RESILIENCE

Summary

Some risks from AI depend on how it is adopted and used. To ensure a safe transition to an AI-enabled society, we need to build resilience into systems and infrastructure, preparing them for the arrival of highly capable models. However, at present, we lack data concerning the nature and extent of AI uptake by individuals and organisations. To understand and address risks that emerge from the deployment of AI, we need to carry out a rigorous and systematic assessment of how AI products and services are being integrated into different sectors, measuring their real-world impacts on individuals and organisations, and forecasting their potential for future harm as AI capabilities grow. Insights from this work can then inform practical mitigations, guidance, and solutions that build resilience into the systems and infrastructure in which frontier AI is being deployed.

Key Research Challenges

Our overall research goals in this area are:

- To build a centralised body of knowledge that tracks the nature and extent of frontier AI uptake by individuals and both public and private sector organisations.
- To rigorously model and forecast the large-scale risks that may ensue from rapid and uncontrolled adoption of frontier AI.
- To design mitigations that will pave the way for safe AI adoption and build resilience across sectors and society to the risks that arise.

Examples of the sorts of large-scale risks that we have in mind include:

- As frontier AI systems become embedded in technical tools, workflows and networks, people have reduced oversight of consequential decisions, so that harmful actions by AI go unchecked.
- As frontier AI systems dramatically reduce the need for human manual and knowledge work, we see significant and disruptive changes to the labour market and economy.
- As frontier AI systems become more capable, they are increasingly used to mount cyberattacks on critical national infrastructure, such as systems for power and telecommunications.
- As multimodal features in frontier AI systems improve, criminal organisations deploy them for scams and financial fraud, especially targeting the more vulnerable.
- As frontier AI systems are increasingly deployed on digital platforms, they are weaponised to pollute reliable sources of human knowledge, degrading the information ecosystem.
- As humans increasingly delegate problem-solving and decision-making to frontier AI systems, people become over-reliant on AI reasoning and knowledge, leading to human disempowerment.
- As frontier AI systems come to exhibit humanlike behaviours and personalised features, people become more vulnerable to persuasion, manipulation or forms of emotional dependence.
- As decisions are increasingly made in interconnected networks of AI systems, unstable or volatile dynamics emerge in highly interoperable environments, such as financial markets.

SOCIETAL RESILIENCE

Potential Research Directions

We are interested in work that explores current and anticipated impacts of frontier AI on critical infrastructure, health, education, labour impacts, law, and finance. For each of these sectors we are interested in funding research that will:

- Clarify and sharpen our threat models using an evidence-based approach.
- Define variables of interest for measurement and monitoring.
- Identify and engage with reliable sources of information to track evolving risks.
- Devise and assess metrics that quantify vulnerability to identified risks, both across society and in specific populations.
- Propose and compare mitigations to the risks identified.

The sorts of variables we might wish to monitor includes:

- The degree of embeddedness of AI agents in enterprise workflows, and the extent to which an unsafe level of reliance on AI is being created.
- The prevalence of frontier AI deployment on digital platforms, and the extent to which people are (potentially unwittingly) interacting with AI on social media or messaging services.
- The nature of human-AI interpersonal interactions, and the degree to which this leaves people vulnerable to influence by frontier AI.
- The extent to which current professional roles are at risk of redundancy as AI is increasingly adopted in the private sector.
- The ways in which frontier AI models are being used by private consumers, and the frequency and longevity of human-AI interactions.
- Incidence of frontier AI being used for fraud and scams, and the features that make this possible.

Policy Impact

The empirical understanding of the risks of deploying frontier AI that we develop through measuring adoption, usage and incidents will advance specific mitigations and inform policy recommendations that build resilience into societal systems and infrastructure.

References to Existing Work

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- Raji et al. (2022) Outsider Oversight: Designing a Third Party Audit Ecosystem for AI Governance. arxiv.org/abs/2206.04737
- Stein et al. (2024) The Role of Governments in Increasing Interconnected Post-Deployment Monitoring of AI. doi.org/10.48550/arXiv.2410.04931
- AISI Systemic Safety grants [page](#)
- AI Policy Perspectives [blog](#)

Contact us

For further questions or clarifications, you may refer to our [Application Pack, Clarification Questions \(CQ\)](#) document or message us at aisichallengefund@dsit.gov.uk.